

Accessibility

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Figure 1: QR Code



Before the Test

Cover Your Bases

- *Always* have a clear, specific reference/citation to where a correct answer is pulled from - to textbook, lecture notes, etc. Consider including this on the test itself
- If the content of lectures/readings changed, ensure questions are subtracted or added
- Review questions before each test administration, even if unchanged since last time

Know Your Audience

- Consider surveying or polling your students about the preferences/anxieties about different types of assignments; consider reasonable modifications
- Give adequate warning and reminders of impending tests
- Have a plan for the, "How do I study?" question; and make sure you believe in it!

Write Good Questions

- Look at stated learning objectives, or big header topics; considered repeated vocabulary; avoid "niche" or "nitpicky" questions
- Avoid using words in questions above 12th grade reading level, unless they were taught as vocabulary
- Ideate using concrete artifacts used in lessons, e.g., readings, notes, worksheets, etc.; avoid coming up with things off the top of your head

After the Test

In general, try to make decisions about individual items, not the test as a whole. Think of each individual question as a test in and of itself.

Understanding Test Item Metrics

“Difficulty”

- Each question has P or the proportion of students that got a question correct, e.g. $P = 0.80$ means 80% of students got a question correct
- Higher $P \rightarrow$ easier question

“Discrimination”

- How well does a question correlate to the test as a whole? Usually given as r_{pbis} or the point-biserial correlation, e.g. $r_{pbis} = 0.60$ means a high degree of correlation with the test
- Higher, positive r_{pbis} greater discrimination

“Subjective Content Relevance”

- Was a question about a central, learning objective-related concept? Or a footnote?
- How many times was this concept covered, and in how many different mediums

Responding to Test Item Metrics

- **Red flag items:** those that are both difficult ($P < 0.30$) and poorly discriminating ($r_{pbis} < 0.20$); examine each with subjective content relevance - does this question capture the student's knowledge in this domain well?
- **Options for responding:**
 - “Dropping” a test item: remove it from the test entirely, but hurts those who got it right; most conservative solution
 - “Nullifying” a test item: giving full credit to several or all possible choices, giving a “freebie” to all students; most generous solution
 - “Curving” the test as a whole: provides all students with a boost, but is primarily focused on a certain average or grade outcome; indifferent to actual question performance
 - Ignore it
- Consider a **test wrapper**: An activity that requests corrections to the test and debriefs feelings around it. This can heal frustrations and give points for engaging with a poor performance rather than ignoring.